JAMES SMITH

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jamesdsmith.info

RESEARCH INTERESTS

Human-Computer Interaction, Mixed Reality, Immersive Computing, Computer Vision, Computational Imaging

EDUCATION

University of California, Berkeley. Berkeley, CA
Ph.D Student in Computer Science **2019 – 2026**

University of California, Berkeley. Berkeley, CA
M.S in Electrical Engineering and Computer Science **2018 – 2019**

University of California, Berkeley. Berkeley, CA
B.S. in Electrical Engineering and Computer Science **2015 – 2018**

Sierra College. Rocklin, CA
Study for transfer in Electrical Engineering **2011 – 2015**

RESEARCH EXPERIENCE

University of California, Berkeley. Berkeley, CA
Graduate Student Researcher *with Björn Hartmann* **Aug 2018 – Aug 2026**

Microsoft Research. Redmond, WA (Remote)
Research Intern *with Andy Wilson* **May 2021 - Aug 2021**

University of California, Berkeley. Berkeley, CA
Undergraduate Student Researcher *with Björn Hartmann* **Jun 2017 – May 2018**

University of California, Berkeley. Berkeley, CA
Undergraduate Student Researcher *with Avideh Zakhor* **Nov 2015 – Dec 2016**

PUBLICATIONS

James Smith, Samia Menon, Jessie Liu, Vivian Chan, Bjoern Hartmann. Predicting Tool Extension in VR Interactions Using Gaze Features. *UIST 2026*. **2026**

James Smith, Shm Garanganao Almeda, Tim Aveni, Anya Agarwal, Bjoern Hartmann. Noise Pilot: Enabling Artistic Workflow Composition with Diffusion-Based Image Generation. *CHI 2026*. ***Best Paper Honorable Mention*** **2026**

Nicholas Jennings, Han Wang, Isabel Li, **James Smith**, Bjoern Hartmann. What's the Game, then? Opportunities and Challenges for Runtime Behavior Generation. *UIST 2024*. ***Best Paper Award*** **2024**

James Smith, Xinyun Cao, Adolfo Ramirez-Aristizabal, Bjoern Hartmann. Dual Body Bimanual Coordination in Immersive Environments. *DIS 2023* **2023**

Ananya Nandy, **James Smith**, Nicholas Jennings, Mike Kuniavsky, Kosa Goucher-Lambert, Björn Hartmann. VR or Not? Investigating Interface Type and User Strategies for Interactive Design Space Exploration. *ICED 2023* **2023**

Nicholas Jennings, Ananya Nandy, Xinyi Zhu, Yuting Wang, Fanping Sui, **James Smith**, Björn Hartmann. GeneratiVR: Spatial Interactions in Virtual Reality to Explore Generative Design Spaces. *CHI 2022 EA* **2022**

Andrew Head, Jason Jiang, **James Smith**, Marti Hearst, Björn Hartmann. Composing Flexibly-Organized Step-by-Step Tutorials from Linked Source Code, Snippets, and Outputs. *CHI 2020*. ***Honorable Mention*** **2020**

Michael Laielli*, **James Smith***, Giscard Biamby*, Trevor Darrell, Björn Hartmann. LabelAR: A Spatial Guidance Interface for Fast Computer Vision Image Collection. *UIST 2019* **2019**

Eldon Schoop, **James Smith**, Björn Hartmann. HindSight: Enhancing Spatial Awareness by Sonifying Detected Objects in Real-Time 360-Degree Video. *CHI 2018* **2018**

WORK EXPERIENCE

Jacobs Institute for Design Innovation. Berkeley, CA
Principle Engineer, Virtual Reality Software **Jan 2019 – Jan 2020**
Prototyping and developing applications and prototypes exploring the usage of virtual reality systems in academic maker spaces.

Sierra College. Rocklin, CA
Math and Physics Tutor **Jan 2014 - May 2015**
Individualized Learning Program - special program for algebra students with math anxiety. Lead tutor for undergraduate lower division physics series.

Sony Online Entertainment. Bellevue, WA
Gameplay Programmer **Jun 2008 - Oct 2010**
Game camera, player movement systems, input management, UI, animation for an online first person shooter game.

Zombie Studios. Seattle, WA
Gameplay Programmer **May 2005 - Jun 2008**
Game camera, player movement systems, UI, networking for first and third person shooter games.

TEACHING

University of California, Berkeley. Berkeley, CA
CS 160 - User Interface Design and Development **Fall 2025**
Instructor Rating: 6.92 / 7 (6.18 dept avg)

CS 160 - User Interface Design and Development **Summer 2025**
Instructor Rating: 6.50 / 7 (6.37 dept avg)

CS 160 - User Interface Design and Development **Spring 2025**
Instructor Rating: 6.56 / 7 (6.28 dept avg)

CS 260B - HCI Research **Fall 2024**
Instructor Rating: 6.60 / 7 (6.28 dept avg)

CS 160 - User Interface Design and Development **Summer 2023**
Instructor Rating: 6.47 / 7 (6.22 dept avg)

CS 294-137 - Immersive Computing and Virtual Reality **Fall 2023**
Instructor Rating: 6.69 / 7 (6.26 dept avg)

EECS 149/249A - Embedded Systems **Fall 2021**
Instructor Rating: 4.83 / 5 *(4.30 dept avg)*
Outstanding GSI Award (Top 10% of campus GSIs)

Sierra College. Rocklin, CA
Peer-Led Team Learning - Physics **Fall 2014 - Spring 2015**

AWARDS AND HONORS

EECS Outstanding TA Award, *UC Berkeley* **2025**
Accenture Research Gift **2021-2024**
Demetri Angelakos Memorial Achievement Award, *UC Berkeley* **2023**
Outstanding Graduate Student Instructor, *UC Berkeley* **2021**
James H. Eaton Memorial Scholarship, *UC Berkeley* **2017**
EECS Undergraduate Department Scholarship, *UC Berkeley* **2017**

IEEE Eta Kappa Nu (HKN) Honor Society, *Mu Chapter, UC Berkeley* **2015**
The Tod and Bev Santos Endowed Undergraduate Scholarship, *UC Berkeley* **2015**
NASA National Community College Aerospace Scholar, *NASA* **2013**

SERVICE

Reviewed Papers At: VRST ’21, DIS ’23, UIST ’23, VRST ’24, CHI ’24, CHI ’25, UIST ’25, CHI ’26
EECS Transfer Student Panel Facilitator, *UC Berkeley* **2018 - 2019**
EECS Department Relations, Community Service Officer, *HKN Mu Chapter* **Jan 2015 - Dec 2017**